SMC

14 May 2023 11:51

1. What do you mean by soft margin. /Linearly -inseparable?
2. How to handle the misclassified points in SMC?
3. What is the objective function of SMC?
4. What is the meaning of regularization and Hinge loss?
5. What are the unknown parameters in the objective function?
6. How to solve a primal convex minimization problem with quadratic inequalities using optimizations techniques?
7. What do you mean by dual, how do you realize it in SMC?
8. What are the values of unknown parameters, obtain it?
9. What is final result of SMC classifier for a new point?

→ $(w^T x_i + b) y_i \geq 1$ (LMCS)

$(w^T x_i + b) \cdot y_i \geq 0$

$\forall$ $(w^T x_i + b) \cdot y_i \geq 1 - \xi_i$ $\;\xi = 0$

$\forall x_i \in D$

→ $\xi = 0$ ⇒ LMC

$0 \leq \xi < 1$ ⇒ correctly classified.

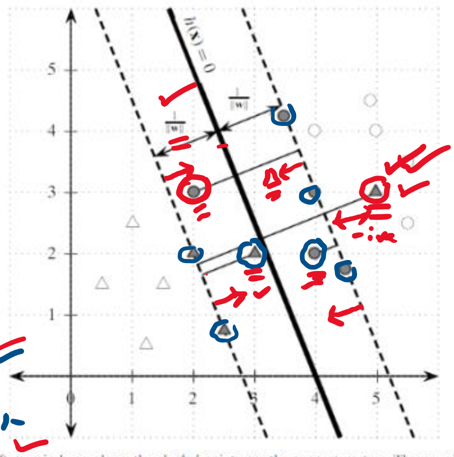

Figure 21.3. Soft margin hyperplane: the shaded points are the support vectors. The margin is $1/\|w\|$ as illustrated, and points with positive slack values are also shown (thin black line).

$\xi \geq 1$ ⇒ -ive value

$1 - \xi$ → Negative

→ objective function :-

$k=1$ → Hinge loss

$k=2$

$$\min_{w,b,\xi} \left(\frac{\|w\|^2}{2}\right) + C \cdot \sum_{i=1}^{n} (\xi_i)^k$$

LM

Regularization constant → No. of errors

Values of each error

Loss

→ $(w^T x_i + b) \cdot y_i \geq 1 - \xi$

$\xi \geq 0$

$C \to 0$

$C \to \infty$

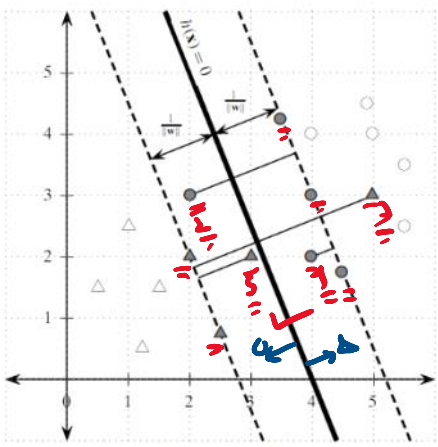

Figure 21.3. Soft margin hyperplane: the shaded points are the support vectors. The margin is $1/\|w\|$ as illustrated, and points with positive slack values are also shown (thin black line).

What is the meaning of regularization and Hinge loss?
What are the unknown parameters in the objective function?
How to solve a primal convex minimization problem with quadratic inequalities using optimizations

21.3.1 Hinge Loss

Assuming $k = 1$, we can compute the Lagrangian for the optimization problem in Eq. (21.19) by introducing Lagrange multipliers α_i and β_i that satisfy the following KKT conditions at the optimal solution:

$$\alpha_i\left(y_i(\mathbf{w}^T\mathbf{x}_i + b) - 1 + \xi_i\right) = 0 \text{ with } \alpha_i \geq 0$$
$$\beta_i(\xi_i - 0) = 0 \text{ with } \beta_i \geq 0 \tag{21.20}$$

$1-\xi_i$, $\alpha_i = 0 \Rightarrow$

The Lagrangian is then given as

1)
$$L = \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{n}\xi_i - \sum_{i=1}^{n}\alpha_i\left(y_i(\mathbf{w}^T\mathbf{x}_i + b) - 1 + \xi_i\right) - \sum_{i=1}^{n}\beta_i\xi_i \tag{21.21}$$

$$\frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}\xi_i - \sum_{i=1}^{n}\alpha_i y_i w^T x_i - \sum_{i=1}^{n}\alpha_i y_i b + \sum_{i=1}^{n}\alpha_i - \sum_{i=1}^{n}\alpha_i\xi_i - \sum_{i=1}^{n}\beta_i\xi_i$$

We turn this into a dual Lagrangian by taking its partial derivative with respect to $\mathbf{w}$, b and ξ_i, and setting those to zero:

2)
$$\frac{\partial}{\partial \mathbf{w}}L = \mathbf{w} - \sum_{i=1}^{n}\alpha_i y_i \mathbf{x}_i = \mathbf{0} \quad \text{or} \quad \mathbf{w} = \sum_{i=1}^{n}\alpha_i y_i \mathbf{x}_i$$

(LMC)

$$\frac{\partial}{\partial b}L = \sum_{i=1}^{n}\alpha_i y_i = 0$$
$$\frac{\partial}{\partial \xi_i}L = C - \alpha_i - \beta_i = 0 \quad \text{or} \quad \beta_i = C - \alpha_i \tag{21.22}$$

$$\frac{\partial w}{\partial w} - \sum_{i=1}^{n}\alpha_i y_i x_i$$

$$\alpha_i \sum_{i=1}^{n}\xi_i$$

$$\beta_i \sum_{i=1}^{n}\xi_i$$

What do you mean by dual, how do you realize it in SMC?

8. What are the values of unknown parameters, obtain it?
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Plugging these values into Eq. (21.21), we get

$$L_{dual} = \frac{1}{2}\mathbf{w}^T\mathbf{w} - \mathbf{w}^T\underbrace{\left(\sum_{i=1}^{n}\alpha_i y_i \mathbf{x}_i\right)}_{\mathbf{w}} - b\underbrace{\sum_{i=1}^{n}\alpha_i y_i}_{0} + \sum_{i=1}^{n}\alpha_i + \sum_{i=1}^{n}\underbrace{(C - \alpha_i - \beta_i)}_{0}\xi_i$$
$$= \sum_{i=1}^{n}\alpha_i - \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n}\alpha_i\alpha_j y_i y_j \mathbf{x}_i^T\mathbf{x}_j$$

$$\Rightarrow \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}\xi_i - \sum_{i=1}^{n}\alpha_i y_i w^T x_i - \sum_{i=1}^{n}\alpha_i y_i b + \sum_{i=1}^{n}\alpha_i - \sum_{i=1}^{n}\alpha_i\xi_i - \sum_{i=1}^{n}\beta_i\xi_i$$

$$\frac{1}{2}w^T w - w^T\sum_{i=1}^{n}\alpha_i y_i x_i - b\sum_{i=1}^{n}\alpha_i y_i + \sum_{i=1}^{n}\alpha_i + C\sum_{i=1}^{n}\xi_i - \alpha_i\sum_{i=1}^{n}\xi_i - \beta_i\sum_{i=1}^{n}\xi_i$$

The dual objective is thus given as

$$\boxed{\begin{array}{l}\textbf{Objective Function: } \max_{\alpha} L_{dual} = \sum_{i=1}^{n}\alpha_i - \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n}\alpha_i\alpha_j y_i y_j \mathbf{x}_i^T\mathbf{x}_j \\ \textbf{Linear Constraints: } 0 \leq \alpha_i \leq C, \ \forall i \in \mathbf{D} \text{ and } \sum_{i=1}^{n}\alpha_i y_i = 0\end{array}} \tag{21.23}$$

$$\frac{1}{2}w^T w - w^T w + \sum_{i=1}^{n}\alpha_i$$
$$= \sum_{i=1}^{n}\alpha_i - \frac{1}{2}w^T w$$

Weight Vector and Bias

Once we solve for α_i, we have the same situation as before, namely, $\alpha_i = 0$ for points that are not support vectors, and $\alpha_i > 0$ only for the support vectors, which comprise all points $\mathbf{x}_i$ for which we have

$y_i((w^T.x_i + b).y - 1 + \xi) = 0$

→ Dell s.v

$(\alpha_i > 0)$

$$y_i(\mathbf{w}^T\mathbf{x}_i + b) = 1 - \xi_i \tag{21.24}$$

Notice that the support vectors now include all points that are on the margin, which have zero slack ($\xi_i = 0$), as well as all points with positive slack ($\xi_i > 0$).

→ $0 < \xi < 1$

We can obtain the weight vector from the support vectors as before:

$$\mathbf{w} = \sum_{\alpha_i > 0} \alpha_i y_i \mathbf{x}_i \tag{21.25}$$

We can also solve for the β_i using Eq. (21.22):

$$\beta_i = C - \alpha_i$$

Replacing β_i in the KKT conditions [Eq. (21.20)] with the expression from above we obtain

$\beta_i \cdot \xi = 0$

$$(C - \alpha_i)\xi_i = 0 \tag{21.26}$$

Thus, for the support vectors with $\alpha_i > 0$, we have two cases to consider:

(1) $\xi_i > 0$, which implies that $C - \alpha_i = 0$, that is, $\alpha_i = C$, or

(2) $C - \alpha_i > 0$, that is $\alpha_i < C$. In this case, from Eq. (21.26) we must have $\xi_i = 0$. In other words, these are precisely those support vectors that are on the margin.

Using those support vectors that are on the margin, that is, have $0 < \alpha_i < C$ and $\xi_i = 0$, we can solve for b_i:

$$\alpha_i \left(y_i(\mathbf{w}^T\mathbf{x}_i + b_i) - 1\right) = 0$$
$$y_i(\mathbf{w}^T\mathbf{x}_i + b_i) = 1$$
$$b_i = \frac{1}{y_i} - \mathbf{w}^T\mathbf{x}_i = y_i - \mathbf{w}^T\mathbf{x}_i \tag{21.27}$$

$\frac{1}{-1}$, $\frac{1}{+1}$

y_i – / -1 , $+$

To obtain the final bias b, we can take the average over all the b_i values. From Eqs. (21.25) and (21.27), both the weight vector $\mathbf{w}$ and the bias term b can be computed without explicitly computing the slack terms ξ_i for each point.

Once the optimal hyperplane plane has been determined, the SVM model predicts the class for a new point $\mathbf{z}$ as follows:

$$\hat{y} = \mathrm{sign}(h(\mathbf{z})) = \mathrm{sign}(\mathbf{w}^T\mathbf{z} + b)$$